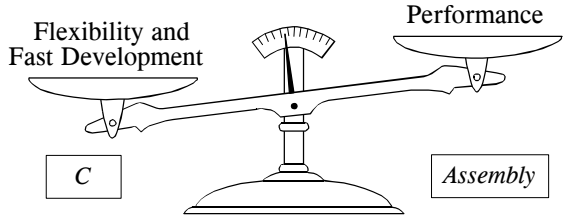

FIGURE 28-10
Assembly versus C. Programs in C are more flexible and quicker to develop. In comparison, programs in assembly often have better performance; they run faster and use less memory, resulting in lower cost.

the hardware if you so desire. Even if you intend to program only in C, you will probably need some knowledge of the architecture of the DSP and the assembly instruction set. For instance, look back at lines 002 and 003 in Table 28-2, the dot product program in C. The "dm" means that *x*[ ] is to be stored in data memory, while the "pm" indicates that *y*[ ] will reside in program memory. Even though the program is written in a high level language, a basic knowledge of the hardware is still required to get the best performance from the device.

Which language is best for *your* application? It depends on what is more important to you. If you need flexibility and fast development, choose C. On the other hand, use assembly if you need the best possible performance. As illustrated in Fig. 28-10, this is a tradeoff you are forced to make. Here are some things you should consider.

- ❑ How complicated is the program? If it is large and intricate, you will probably want to use C. If it is small and simple, assembly may be a good choice.
- ❑ Are you pushing the maximum speed of the DSP? If so, assembly will give you the last drop of performance from the device. For less demanding applications, assembly has little advantage, and you should consider using C.
- ❑ How many programmers will be working together? If the project is large enough for more than one programmer, lean toward C and use in-line assembly only for time critical segments.
- ❑ Which is more important, *product cost* or *development cost*? If it is product cost, choose assembly; if it is development cost, choose C.
- ❑ What is your background? If you are experienced in assembly (on other microprocessors), choose assembly for your DSP. If your previous work is in C, choose C for your DSP.
- ❑ What does the DSP's manufacturer suggest you use?

This last item is very important. Suppose you ask a DSP manufacturer which language to use, and they tell you: *"Either C or assembly can be used, but we*